UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

STAT 576: Empirical Process Theory and Weak Convergence Fall 2025
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Lecture 18

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1. Symmetrization for 1D-Thresholds
2. Glivenko-Cantelli Theorem

1 Symmetrization for 1D-Thresholds

1.1 Symmetrization Lemma

Lemma 1. *Let $\mathcal{F}$ be a class of real-valued functions on a measurable space $\mathcal{X}$, $X_1, \dots, X_n$ be i.i.d. random variables taking values in $\mathcal{X}$, and let $\varepsilon_1, \dots, \varepsilon_n$ be independent Rademacher random variables (i.e., $\mathbb{P}(\varepsilon_i = 1) = \mathbb{P}(\varepsilon_i = -1) = \frac{1}{2}$), independent of $(X_i)_{i=1}^n$, Then,*

$$\mathbb{E}\left[\sup_{f\in\mathcal{F}}\left(\mathbb{E}[f(X)] - \frac{1}{n}\sum_{i=1}^n f(X_i)\right)\right] \le 2\,\mathbb{E}\left[\sup_{f\in\mathcal{F}}\frac{1}{n}\sum_{i=1}^n \varepsilon_i f(X_i)\right]$$

Remark. *The quantity $\frac{1}{n}\sum_{i=1}^n \varepsilon_i f(X_i)$ can be interpreted as the correlation between $(f(X_i))_{i=1}^n$ and randomly chosen noise $(\varepsilon_i)_{i=1}^n$. The proof of this lemma is based on the idea of **symmetrization**:*

If $X \overset{\text{i.i.d.}}{\sim} X'$, then $X - X' \overset{d}{=} X' - X \overset{d}{=} \varepsilon(X - X')$, where $\varepsilon \sim$ Rad$(\frac{1}{2})$ and independent from X, X'.

Proof. We introduce i.i.d. copies $(X_i')_{i=1}^n$ such that $X_i' \overset{d}{=} X_i$ for all i. Note that

$$\mathbb{E}[f(X)] = \frac{1}{n}\sum_{i=1}^n \mathbb{E}[f(X_i)] = \frac{1}{n}\sum_{i=1}^n \mathbb{E}[f(X_i')],$$

and therefore,

$$
\begin{aligned}
&\mathbb{E}_X\left[\sup_{f\in\mathcal{F}}\Big(\mathbb{E}[f(X)]-\frac{1}{n}\sum_{i=1}^n f(X_i)\Big)\right]\\
&=\mathbb{E}_X\left[\sup_{f\in\mathcal{F}}\Big(\frac{1}{n}\sum_{i=1}^n \mathbb{E}_{X'}[f(X_i')]-\frac{1}{n}\sum_{i=1}^n f(X_i)\Big)\right]\\
&=\mathbb{E}_X\left[\sup_{f\in\mathcal{F}}\mathbb{E}_{X'}\left[\frac{1}{n}\sum_{i=1}^n\big(f(X_i')-f(X_i)\big)\right]\right]\\
&\le\mathbb{E}_X\left[\mathbb{E}_{X'}\left[\sup_{f\in\mathcal{F}}\frac{1}{n}\sum_{i=1}^n\big(f(X_i')-f(X_i)\big)\right]\right]\quad\text{(by Jensen's inequality)}\\
&=\mathbb{E}_{X,X',\varepsilon}\left[\sup_{f\in\mathcal{F}}\frac{1}{n}\sum_{i=1}^n\varepsilon_i\big(f(X_i')-f(X_i)\big)\right]\quad\text{(by symmetrization)}\\
&\le\mathbb{E}_{X,X',\varepsilon}\left[\sup_{f\in\mathcal{F}}\frac{1}{n}\sum_{i=1}^n\varepsilon_i f(X_i')\right]+\mathbb{E}_{X,X',\varepsilon}\left[\sup_{f\in\mathcal{F}}\frac{1}{n}\sum_{i=1}^n(-\varepsilon_i f(X_i))\right]\quad\text{(by the supremum property)}\\
&=2\,\mathbb{E}_{X,\varepsilon}\left[\sup_{f\in\mathcal{F}}\frac{1}{n}\sum_{i=1}^n\varepsilon_i f(X_i)\right],\quad\text{(since }\varepsilon_i f(X_i')\overset{d}{=}-\varepsilon_i f(X_i)\overset{d}{=}\varepsilon_i f(X_i))
\end{aligned}
$$

□

Lemma 2. *Using the same notations as above, we have*

$$
\mathbb{E}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^n\varepsilon_i f(X_i)\right|\right]\le 2\,\mathbb{E}\left[\sup_{f\in\mathcal{F}}\left|\mathbb{E}[f(X)]-\frac{1}{n}\sum_{i=1}^n f(X_i)\right|\right]+\frac{c}{\sqrt{n}}\sup_{f\in\mathcal{F}}\big|\mathbb{E}[f(X)]\big|
$$

Proof. Similarly, we construct i.i.d. copies $X_1',\dots,X_n'$ such that $X_i'\overset{d}{=}X_i$ for all i, and note that $\mathbb{E}[f(X)]=\mathbb{E}[f(X_i)]=\mathbb{E}[f(X_i')]$. Then,

$$
\begin{aligned}
\mathbb{E}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^n\varepsilon_i f(X_i)\right|\right]&=\mathbb{E}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^n\varepsilon_i\big(f(X_i)-\mathbb{E}[f(X_i)]\big)+\frac{1}{n}\sum_{i=1}^n\varepsilon_i\mathbb{E}[f(X_i)]\right|\right]\\
&\le\underbrace{\mathbb{E}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^n\varepsilon_i\big(f(X_i)-\mathbb{E}[f(X_i)]\big)\right|\right]}_{(1)}+\underbrace{\mathbb{E}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^n\varepsilon_i\mathbb{E}[f(X)]\right|\right]}_{(2)}.
\end{aligned}
$$

For term (1), we replace $\mathbb{E}_X[f(X_i)]$ by $\mathbb{E}_{X'}[f(X_i')]$ since they are identically distributed:

$$
\begin{aligned}
(1) &= \mathbb{E}_X\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^n \varepsilon_i\big(f(X_i)-\mathbb{E}_{X'}[f(X_i')]\big)\right|\right] \\
&\le \mathbb{E}_{X,X',\varepsilon}\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^n \varepsilon_i\big(f(X_i)-f(X_i')\big)\right| \quad \text{(by Jensen's inequality)} \\
&= \mathbb{E}_{X,X'}\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^n \big(f(X_i)-f(X_i')\big)\right| \quad \text{(by symmetrization)} \\
&= \mathbb{E}_{X,X'}\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^n \big(f(X_i)-\mathbb{E}[f(X)]+\mathbb{E}[f(X)]-f(X_i')\big)\right| \\
&\le 2\,\mathbb{E}_{X,X'}\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^n \big(f(X_i)-\mathbb{E}[f(X)]\big)\right| \quad \text{(by the triangle inequality).}
\end{aligned}
$$

For term (2), we can upper-bound it as follows:

$$
\begin{aligned}
(2) &= \sup_{f\in\mathcal{F}}|\mathbb{E}[f(X)]|\cdot\mathbb{E}_\varepsilon\left[\left|\frac{1}{n}\sum_{i=1}^n \varepsilon_i\frac{\mathbb{E}[f(X)]}{|\mathbb{E}[f(X)]|}\right|\right] \quad \text{(since } \mathbb{E}[f(X)] \text{ is a constant)} \\
&= \sup_{f\in\mathcal{F}}\mathbb{E}_\varepsilon\left[\left|\frac{1}{n}\sum_{i=1}^n \varepsilon_i\right|\right]\cdot|\mathbb{E}[f(X)]| \quad \text{(since } \varepsilon_i \overset{d}{=} \varepsilon_i\tfrac{\mathbb{E}[f(X)]}{|\mathbb{E}[f(X)]|}\text{)} \\
&\le \frac{1}{\sqrt{n}}\sup_{f\in\mathcal{F}}|\mathbb{E}[f(X)]| \quad \text{(by Jensen's inequality and } \mathbb{E}_\varepsilon\Big|\sum_{i=1}^n\varepsilon_i\Big| \le \sqrt{\mathbb{E}_\varepsilon\big(\sum_{i=1}^n\varepsilon_i\big)^2} = \sqrt{n}\text{).}
\end{aligned}
$$

In fact, it indicates that $c = 1$ in above lemma 2. □

1.2 1D-Thresholds

Recall that we aim to control

$$\mathbb{E}\left[\sup_{\theta\in\mathbb{R}}\left(\mathbb{E}[\mathbb{1}(X\le\theta)]-\frac{1}{n}\sum_{i=1}^n\mathbb{1}(X_i\le\theta)\right)\right]$$

By Lemma 1 (symmetrization), we obtain the upper bound

$$\mathbb{E}\left[\sup_{\theta\in\mathbb{R}}\left(\mathbb{E}[\mathbb{1}(X\le\theta)]-\frac{1}{n}\sum_{i=1}^{n}\mathbb{1}(X_i\le\theta)\right)\right] \;\le\; 2\,\mathbb{E}\left[\sup_{\theta\in\mathbb{R}}\left(\frac{1}{n}\sum_{i=1}^{n}\varepsilon_i\mathbb{1}(X_i\le\theta)\right)\right]$$

Denote

$$V_\theta=\frac{1}{n}\sum_{i=1}^{n}\varepsilon_i\,\mathbb{1}(X_i\le\theta),\qquad \theta\in\mathbb{R}.$$

If we order the sample such that $X_1<X_2<\cdots<X_n$, then V_θ changes only at the observed sample points. Hence, V_θ takes at most $n+1$ distinct values corresponding to these jump points.

$$\begin{aligned}\mathbb{E}\left[\sup_{\theta\in\mathbb{R}}\frac{1}{n}\sum_{i=1}^{n}\varepsilon_i\mathbb{1}(X_i\le\theta)\right] &= \mathbb{E}_X\left[\mathbb{E}_\varepsilon\left[\sup_{\theta\in\mathbb{R}}V_\theta\right]\;\middle|\;X\right]\\ &= \mathbb{E}_X\left[\mathbb{E}_\varepsilon\left[\sup_{\theta\in\{\theta_1,\dots,\theta_{n+1}\}}V_\theta\right]\;\middle|\;X\right]\\ &\le \frac{1}{\sqrt{n}}\sqrt{2\log(n+1)}\;=\;O\left(\sqrt{\frac{\log n}{n}}\right)\end{aligned}$$

where the last inequality follows since V_θ is $\frac{1}{\sqrt{n}}$-sub-Gaussian by maximal inequalities.

2 Glivenko-Cantelli Theorem

Definition 1 (Rademacher Complexity)**.** *Given a class $\mathcal{F}$ of functions, the **Rademacher Complexity** $\mathfrak{R}(\mathcal{F})$ is given by:*

$$\mathfrak{R}(\mathcal{F})=\mathbb{E}_{X,\varepsilon}\sup_{f\in\mathcal{F}}\Big|\frac{1}{n}\sum_{i=1}^{n}\varepsilon_i f(X_i)\Big|$$

Remark. *If we replace ε_i with $g_i\sim\mathcal{N}(0,1)$, then we get **Gaussian Complexity**.*

Definition 2 (Glivenko-Contelli Class)**.** *Given a class $\mathcal{F}$ of measurable functions, define:*

$$\|\mathbb{P}_n-\mathbb{P}\|_{\mathcal{F}}:=\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^{n}f(X_i)-\mathbb{E}[f(X)]\right|$$

The class $\mathcal{F}$ is said to be a Glivenko–Cantelli class *if*

$$\|\mathbb{P}_n - \mathbb{P}\|_{\mathcal{F}} \xrightarrow{p} 0$$

Example : $\mathcal{F} = \{x \to \mathbb{1}(x \leq \theta) : \theta \in \mathbb{R}\}$ is a GC-class.
Example : Not all classes are Glivenko–Cantelli. Suppose that X has a continuous distribution, and consider

$$\mathcal{S} = \{A \subset \mathbb{R} : \mathrm{card}(A) < \infty\}, \qquad \mathcal{F} = \{\mathbb{1}_A : A \in \mathcal{S}\}.$$

Then we claim that

$$\sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^{n} f(X_i) - \mathbb{E}[f(X)] \right| = 1.$$

Since X has a continuous distribution, the probability that it takes any value from a finite set is zero. Indeed,

$$f = \mathbb{1}_{\{X_1, \ldots, X_n\}} \quad \Rightarrow \quad \frac{1}{n} \sum_{i=1}^{n} f(X_i) = 1$$

$$\mathbb{E}[f(X)] = \mathbb{P}(X \in \{X_1, \ldots, X_n\}) = 0$$

And therefore $\mathcal{F}$ is *not* a Glivenko–Cantelli class.

Theorem 1. *Let $\mathcal{F}$ be a b-uniformly bounded class of functions, i.e.,*

$$\sup_{f \in \mathcal{F}} \|f\|_\infty = \sup_{f \in \mathcal{F}} \sup_{x \in \mathbb{R}} |f(x)| \leq b$$

Then, for any $n \geq 1$ and any $\delta > 0$, we have

$$\|\mathbb{P}_n - \mathbb{P}\|_{\mathcal{F}} \;\leq\; 2\,\mathfrak{R}_n(\mathcal{F}) + \delta$$

with probability at least

$$1 - \exp\left(-\frac{n\delta^2}{2b^2}\right)$$

In particular, if $\mathfrak{R}_n(\mathcal{F}) = o(1)$, then

$$\|\mathbb{P}_n - \mathbb{P}\|_{\mathcal{F}} \xrightarrow{a.s.} 0$$

Remark. *To establish the Glivenko–Cantelli property, it suffices to control the Rademacher complexity.*

Proof. Let us define the centered version of each function:

$$\bar{f}(x) = f(x) - \mathbb{E}[f(X)].$$

Then,

$$\|\mathbb{P}_n - \mathbb{P}\|_{\mathcal{F}} = \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^{n} \bar{f}(X_i) \right|.$$

From the symmetrization lemma,

$$\mathbb{E}\left[\sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^{n} \big(f(X_i) - \mathbb{E}[f(X)] \big) \right| \right] \leq 2\, \mathbb{E}\left[\sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^{n} \varepsilon_i f(X_i) \right| \right] = 2\, \mathfrak{R}_n(\mathcal{F}).$$

Hence,

$$\mathbb{E}\big[\|\mathbb{P}_n - \mathbb{P}\|_{\mathcal{F}} \big] \;\leq\; 2\, \mathfrak{R}_n(\mathcal{F}).$$

We have shown the expectation is bounded by Rademacher complexity, then it remains to establish the concentration of $\|\mathbb{P}_n - \mathbb{P}\|_{\mathcal{F}}$ around its mean.
Define

$$G(X_1, \ldots, X_n) = \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^{n} \bar{f}(X_i) \right|.$$

Let us evaluate how G changes when one sample X_i is replaced by an independent copy X_i':

$$\begin{aligned} \big| G(X_1, \ldots, X_n) - G(X_1, \ldots, X_i', \ldots, X_n) \big| &\leq \frac{1}{n} \left| \bar{f}(X_i) - \bar{f}(X_i') \right| \\ &\leq \frac{2b}{n} \quad \text{(b-uniformly bounded class)} \end{aligned}$$

Therefore, **McDiarmid's inequality** then yields the concentration

$$\Pr\big(\big| G(X_1, \ldots, X_n) - \mathbb{E}[G(X_1, \ldots, X_n)] \big| \geq \delta \big) \leq 2 \exp\left(-\frac{n\delta^2}{2b^2} \right).$$

□